

Loop invariant of lines 15-19: Just before the start of the i th iteration of the loop, for all indices K , where $1 \leq K \leq i$, numTerms , the entry $\langle a[K].\text{row}, a[K].\text{col}, a[K].\text{value} \rangle$ of matrix $a[]$ has been correctly transposed to the ^{corresponding unique} index K' in $b[]$, where $b[K'].\text{row} = a[K].\text{col}$, $b[K'].\text{col} = a[K].\text{row}$ and $b[K'].\text{value} = a[K].\text{value}$. The entries in $b[]$ are such that $\forall t_1, t_2, 1 \leq t_1, t_2 \leq \text{current_b}$ (and also, t_1 and t_2 are distinct filled indices) $b[t_1].\text{row} \leq b[t_2].\text{row}$, i.e. the rows are in ascending order of index. Also, if $b[t_1].\text{row} = b[t_2].\text{row}$, then $b[t_1].\text{col} < b[t_2].\text{col}$, i.e. the elements of the same row are arranged in ascending order of column index. Here, current_b denotes the highest index filled in $b[]$ so far.

Initialization: Just before the 1st iteration of the loop starts, i is initialized to 1. The set of elements $1 \leq K \leq i = 1$ is empty. Any claim on an empty set is vacuously true. $\therefore$ Loop invariant holds

Maintenance: Let the loop invariant hold just before the start of the iteration, when $i = x$, $1 \leq x \leq \text{numTerms}$. We know, $a[x].\text{col}$ returns the column index of the term at index x of matrix $a[]$ (By defn.). Let $a[x].\text{col} = c$, $0 \leq c < \text{numCols}$. By correctness of lines 12-14, $\text{startingPos}[c]$ initially stores the index in array $b[]$ of the element having column c in array $a[]$ ($\because$ row c in array $b[]$ by defn. of transpose) and minimum indexed row of all other non-zero elements of $a[]$ having column K (by defn. of transpose, minimum indexed column of all other non-zero elements of $b[]$ having row K).

By correctness of lines 8-11, $\text{rowTerms}[c]$ stores the no. of non-zero elements of $a[]$ having column index c (or, by the defn. of the transpose matrix, the no. of non-zero elements of $b[]$ having row index c).